

# Networks: supply and demand

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*Solution of practice quiz*

1. (a) Remember that the divergence of a flow vector  $x \in \mathbb{R}^n$  at node  $i$  is defined as the total quantity of flow that leaves the node minus the total quantity of flow that enters the node:

$$\operatorname{div}(x)_i = \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki}.$$

If the divergence of node  $i$  is positive, then units of flow are produced at  $i$  and it is a *supply node*. If the divergence of node  $i$  is negative, then units of flow are consumed at  $i$  and it is a *demand node*. If the divergence is zero, it is a *transit node*.

Here we have:

- $\operatorname{div}(x)_{v_0} = (-3) - (5) = -8$ , therefore  $v_0$  is a demand node.
- $\operatorname{div}(x)_{v_1} = (5) - (-3) = 8$ , therefore  $v_1$  is a supply node.
- $\operatorname{div}(x)_{v_2} = 0$ , therefore  $v_2$  is a transit node.
- $\operatorname{div}(x)_{v_3} = (-3) - (6) = -9$ , therefore  $v_3$  is a demand node.
- $\operatorname{div}(x)_{v_4} = 0 - (-3) = 3$ , therefore  $v_4$  is a supply node.
- $\operatorname{div}(x)_{v_5} = (6 + 2) - (4) = 4$ , therefore  $v_5$  is a supply node.
- $\operatorname{div}(x)_{v_6} = (4 - 5) - 0 = -1$ , therefore  $v_6$  is a demand node.
- $\operatorname{div}(x)_{v_7} = (-3) - (2 + 1) = -6$ , therefore  $v_7$  is a demand node.
- $\operatorname{div}(x)_{v_8} = 1 - (-3 - 5) = 9$ , therefore  $v_8$  is a supply node.

(b) We use the results from the previous step to verify that

$$\sum_{i=0}^8 \text{div}(x)_{v_i} = -8 + 8 + 0 - 9 + 3 + 4 - 1 - 6 + 9 = 0.$$

2. For each node  $i \in \mathcal{N}$ , the divergence of  $x$  at node  $i$  is defined as

$$\text{div}(x)_i = \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki}.$$

Therefore, we have

$$\sum_{i \in \mathcal{N}} \text{div}(x)_i = \sum_{i \in \mathcal{N}} \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{i \in \mathcal{N}} \sum_{k|(k,i) \in \mathcal{A}} x_{ki}.$$

The first term involves a sum over all outgoing arcs of all nodes. It is actually a sum over all the arcs in the network. Similarly, the second involves a sum over all ingoing arcs of all nodes. It is also a sum over all the arcs in the network. Therefore, we have

$$\begin{aligned} \sum_{i \in \mathcal{N}} \text{div}(x)_i &= \sum_{i \in \mathcal{N}} \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{i \in \mathcal{N}} \sum_{k|(k,i) \in \mathcal{A}} x_{ki} \\ &= \sum_{(i,j) \in \mathcal{A}} x_{ij} - \sum_{(k,i) \in \mathcal{A}} x_{ki} \\ &= 0. \end{aligned}$$

Intuitively, it means that every unit of flow that is generated somewhere is consumed somewhere else.